What if income shifts from labor to capital? The fiscal and distributional incidence of AI-driven growth in the United States*

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Abstract

I study how AI-driven growth would move United States government revenue, benefit outlays, poverty and inequality, by running two experiment families through PolicyEngine-US, an open-source static microsimulation of the full federal and state tax and transfer system, on openly published microdata. A mechanism sweep shifts 0–100% of labor income into capital income under current law; a scenario set adopts the Budget Lab's calibration of the Karger et al. (2026) forecaster survey — Slow, Moderate and Rapid adoption crossed with three wage-inequality variants — so that remaining differences between the two models are differences in coverage, not assumptions. In Rapid, the forecast tilt toward capital keeps 58% of the revenue gain that shares-fixed growth would deliver on the federal basis comparable to the Budget Lab's (they get 56%), and 47% on the all-government household basis, which adds states and transfers but has no corporate wedge (+272B against +576B); it also erodes most of growth's poverty gain (-1.09pp becomes -0.17pp). Across the wage-inequality variants, net revenue stays within +233B–+313B while the poverty outcome flips sign, from -1.64pp to +1.59pp; automatic transfer responses absorb 35% of the gross-tax swing across those variants, $28B of it in programs outside a tax-only model, and state tax nearly doubles from +30B to +54B. If none of the incremental capital income reaches a tax return, the household sector's Rapid gain falls from +272B to +26B. Reconciled cell by cell against the Budget Lab's published grid, payroll responses agree in all nine as-forecast cells to within $5.1B, while the bridged headline is +344B against their +246B (1.39×), concentrated in the income-tax response to the capital shock.

Keywords: artificial intelligence, factor shares, microsimulation, taxation, transfers, poverty, inequality, state taxes

JEL codes: H24, H53, H71, D31, I32, O33, C63

*Acknowledgements: the cell-level reconciliation in Section 6 is possible because the Budget Lab at Yale publishes its code and complete result grids alongside its reports; I thank them for that practice. All errors are mine. An interactive companion to this paper is at policyengine.org/us/ai-inequality/income-shift; code and data references are in Appendix C.

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Revision (October 2026). The Budget Lab revised Iselin and Nunn (2026) on 9 October 2026. As first published, their report took GDP growth from the economist respondents of the Karger et al. (2026) survey but the 2030 labor shares from all respondents pooled; the revision takes both from the economists, raising the Moderate labor share from 53.8% to 54.0% and the Rapid share from 51.3% to 52.0%. It also replaces CBO's rounded 2030 ratios, which moves their corporate anchor from $486.1B to $477.3B. This version reruns every calibrated scenario on the same model and data with the revised inputs, takes the CBO growth factor from the path their report states rather than from a rounded inversion of their July table, and reconciles against their revised result grid. Rapid capital income now grows 15.60% above baseline rather than 17.28%, and Rapid labor income changes by +0.41% rather than -0.94%. Rapid / proportional net revenue moves from +211B to +272B, the household keep-rate from 37% to 47%, and its poverty change from +0.11 pp to -0.17 pp: the tilt now erodes growth's poverty gain rather than reversing it. With Rapid labor income growing, the household sector also stays in surplus at every realized share of the new capital income (+26B at 0%, where the July inputs gave -60B and a breakeven near 22.5%). The bridged Rapid / expansive headline moves from +301B to +344B, against their +216B and +246B. Slow and the shares-fixed counterfactuals move by at most $0.04B. The September correction note below describes the run before this revision.

Correction (September 2026). Earlier versions of this paper counted Head Start and Early Head Start in household net income. policyengine-us 1.764.6 values both programs at their per-enrollee cost. The microdata carry a modeled Head Start take-up flag but none for Early Head Start, so every eligible child under three and every eligible pregnant person counted as enrolled in Early Head Start. That put $124B of Early Head Start and $5B of Head Start in 2030 baseline net income, rising and falling with the wage shocks because eligibility is income-tested. This version reruns every calibrated scenario on the same model and data with both programs kept out of net income, as current PolicyEngine does. Neither program counts toward SPM resources, so the poverty results are unchanged. Every tax cell is also unchanged. Net revenue cells move by at most $8B, net Gini levels rise by about 0.005 (baseline 0.4625 to 0.4678), and transfers absorb 35% of the Rapid gross-tax swing ($43B), where earlier versions reported 47% ($57B). The mechanism sweep of Section 4.1 has not been rerun and still counts both programs in benefit outlays and net income. This version also corrects the explanation of why the household keep-rate sits below the federal one: the gap is mainly the Budget Lab's corporate wedge on the federal basis, not state taxes and transfers (Section 6).

1 Introduction

Forecasters expect artificial intelligence to raise output and, at the same time, to tilt its composition away from labor and toward capital (Karger et al., 2026). The United States taxes those two kinds of income very differently, and it runs a transfer system that responds automatically when wages fall. So the fiscal and household consequences of AI-driven growth depend not mainly on how much growth arrives, but on which incomes it arrives in — and on which parts of the fiscal system a model can see. The Budget Lab at Yale recently scored AI growth scenarios through a federal individual income and payroll tax microsimulation and found that a growing capital share mutes the revenue gains (Iselin and Nunn, 2026). Their model, by design, stops at the tax code: no transfer programs beyond tax credits, no poverty measurement, no state systems.

This paper runs the same question through the whole system. PolicyEngine-US is an open-source static microsimulation of United States federal and state tax and benefit law — income and payroll taxes, refundable credits, SNAP, SSI, TANF, WIC and the other major transfer programs, and the Supplemental Poverty Measure (SPM) — running on openly published microdata. I use it in two complementary designs. The first is a mechanism sweep: shift 0–100% of positive labor income into capital income, holding modelled market income fixed, and record what current law does at each step. The second is a calibrated scenario set that deliberately adopts the Budget Lab's implementation of the Karger et al. (2026) forecaster survey — Slow, Moderate and Rapid adoption paths crossed with compressive, proportional and expansive wage-inequality variants, plus their shares-fixed counterfactuals — so that where the two models differ, the difference is model coverage rather than scenario assumptions. Because they publish their code and their complete result grid, the comparison can then be made cell by cell on an identical accounting basis.

Four findings organize the paper. First, the composition tilt matters about as much as the growth rate for revenue, and more for poverty. Rapid adoption with factor shares held at today's levels raises net government revenue by +576B in 2030 and cuts SPM poverty by -1.09pp; the same growth with the forecast tilt keeps 47% of that revenue (+272B) and only 16% of the poverty gain (-0.17pp). The tilt's revenue cost, $304B, is about what moving from Slow to Rapid growth adds as forecast ($265B); its 0.92 pp poverty cost is several times the 0.21 pp that the same move in growth makes. The keep-rate is basis-dependent, and the paper reports it both ways: on the narrower federal basis that the Budget Lab measures, the tilt keeps 58% here and 56% in their grid. The household figure is lower mainly because that basis has no corporate wedge, which grows with the capital shock; adding state taxes and the transfer system raises the tilt's dollar cost but barely moves the keep-rate.

Second, the assumption with the least evidence behind it — whether AI compresses or widens the wage distribution, a question the forecaster survey does not elicit and the Budget Lab spans with stylized variants — decides the household verdict while leaving the revenue verdict intact. Across the Budget Lab's own compressive-to-expansive span, Rapid net revenue stays between +233B and +313B, always positive, while the poverty outcome flips sign: -1.64pp under compression, +1.59pp under expansion, a 3.23pp span.

Third, the parts of the fiscal system outside a tax-only model respond materially, and in both directions. Across those same wage-spread variants, gross tax collections swing $123B, and automatic transfer responses absorb 35% of that swing — $28B of it in non-credit programs such as SNAP and SSI that a tax microsimulation does not carry. State tax codes add +42B under proportional Rapid growth and nearly double across the variants, from +30B to +54B. Whether these channels matter is exactly the kind of question a full-system model exists to answer.

Fourth, how much of the new capital income reaches a tax return moves the Rapid revenue answer nearly as much as the choice between Slow and Rapid adoption does ($245B against $265B). If AI's incremental capital income goes unrealized — retained, accrued, or sheltered inside retirement accounts — revenue falls roughly linearly, from +272B at full realization to +26B when none of it is realized; poverty, meanwhile, moves by 0.07 pp across the whole sweep. And whether taxable retirement distributions count as capital income cuts the Rapid estimate by about a third when they are excluded (+272B to +177B).

The reconciliation with the Budget Lab in Section 6 is, to my knowledge, the first cell-level comparison of two independent microsimulations of AI fiscal scenarios. It is reassuring where the models share machinery — payroll responses agree in all nine cells to within $5.1B,

two independent wage distributions meeting the same taxable maximum — and informative where they do not: the bridged headline is +344B against their +246B (1.39×), with the gap concentrated in the income-tax response to the capital shock — nearly constant in absolute terms across the wage-spread cells — consistent with this paper's 17% larger realized-capital base and more top-concentrated allocation rule.

The paper proceeds as follows. Section 2 places the exercise in the literature. Section 3 describes the model, data, and both experiment designs. Section 4 reports the sweep and scenario results, including the transfer-system and state channels. Section 5 reports the realization and capital-scope sensitivities and stability across data builds. Section 6 reconciles with the Budget Lab cell by cell. Section 7 discusses limitations and next steps.

2 Related work

The paper sits at the intersection of three literatures: macro-fiscal analyses of AI, tax-benefit microsimulation of technology shocks, and the factor-share literature that motivates the experiment's central mechanism.

AI and the public finances. Iselin and Nunn (2026) is the direct antecedent: they translate the Karger et al. (2026) forecaster survey into 2030 growth and factor-share scenarios, add stylized wage-distribution variants, and score them through a federal individual income and payroll tax microsimulation built on the IRS Public Use File, with a corporate revenue wedge outside the microsimulation. Their central finding — that growth accruing to capital raises less revenue than the same growth at fixed factor shares, by nearly a factor of two in the Rapid scenario — is the claim this paper re-derives on independent machinery and extends to transfers, poverty and states. Karger et al. (2026) elicit forecasts from economists, AI-company staff, policy researchers, superforecasters and the public; the median respondent expects modest GDP acceleration by 2030, a declining labor share, and rising wealth inequality, with wide disagreement across groups about outcomes under rapid progress. The survey does not elicit a wage-distribution outcome; the compressive-to-expansive axis is the Budget Lab's stylization of that open question, which this paper propagates rather than resolves. Acemoglu (2025) argues on task-level grounds for modest aggregate effects; Bastani and Waldenström (2024) survey the tax-design questions AI raises, including whether existing bases can reach AI-driven rents.

Microsimulation of AI shocks. Closest in method are two companion studies on European systems. Doorley et al. (2026) run scenario-based AI employment, wage and capital-income shocks through the Irish SWITCH model, and Ahmadi (2026) runs occupation-level exposure-based displacement, wage and capital shocks through PolicyEngine UK, treating the incidence of displacement as an explicit scenario axis. This paper is the United States member of that family, but it differs in the shock design: rather than displacement incidence, it varies the factor composition and wage-distribution shape of income growth, matching the Budget Lab's calibration to enable the cell-level reconciliation of Section 6. Occupation-level exposure measures (Felten et al., 2021; Eloundou et al., 2024) are the natural next refinement of the wage channel here, as Section 7 discusses; both this paper's repository and the Budget Lab's roadmap point the same direction.

Factor shares and automation. The premise that technology can move income between labor and capital at scale is not hypothetical: the global decline of the labor share is documented by Karabarbounis and Neiman (2014), and Acemoglu and Restrepo (2018) provide the task-model mechanics by which automation reallocates value added from labor to capital. What

microsimulation adds to that literature is the statutory detail: a dollar moved from wages to capital income changes its withholding, its marginal rate schedule, its payroll treatment, its state treatment, and the transfer payments of the household it leaves — none of which aggregate models carry. The scenarios here take the CBO baseline as the no-AI counterfactual, using the same February 2026 projections the Budget Lab anchors to (Congressional Budget Office, 2026).

3 Model, data, and experiment designs

3.1 Model and data

All results come from PolicyEngine-US version 1.764.6 (PolicyEngine, 2026a), an open-source static microsimulation of United States federal and state tax and benefit law, executed through the `policyengine.py` runtime (version 5.0.1), which certifies the pairing of model version and dataset build. The 2030 baseline is current law as of that model version, including the 2025 reconciliation act. The dataset is `populace-us` build `cae8640` (28 July 2026) (PolicyEngine, 2026b), an openly published enhanced-CPS microdata file for 2024, uprated by the model to the analysis years; scenario growth is defined in excess of the CBO baseline path, which is exactly right only where the model's own uprating tracks CBO's projection. The mechanism sweep of Section 4.1 is run at 2026; the calibrated scenarios are run at 2030, matching the Budget Lab's horizon. In the 2030 baseline, SPM poverty is 12.13% and the household net-income Gini is 0.4678 (household-level, unequivalized; Section 7 discusses comparability).

The model covers federal individual income tax (including capital gains rates, the net investment income tax and the alternative minimum tax), payroll taxes on both employee and employer sides plus self-employment tax, refundable credits (EITC, refundable CTC and others), the major means-tested transfer programs (SNAP, SSI, TANF, WIC, school meals, housing subsidies, broadband subsidies), and the tax and benefit codes of all fifty states and DC. Health programs (Medicaid, CHIP, ACA subsidies) are valued in the model but excluded from net-income accounting in these runs by the model's default simulation switch, so no result here contains a health-program response in either direction. Head Start and Early Head Start are likewise valued in the model but kept out of net income in the calibrated scenarios, as current PolicyEngine does. The mechanism sweep predates that correction and still counts both programs in benefit outlays and net income (see the correction before the introduction). Everything is static: no labor supply, behavioral, avoidance or enforcement responses, matching the Budget Lab's individual microsimulation.

Accounting. Net government revenue is defined as taxes minus transfers: household tax before refundable credits (federal and state, including employee-side payroll) plus employer payroll tax, minus refundable credits and benefit outlays. An identity check — the change in market income minus the change in household net income must equal the household-side revenue change — closes to within $0.001B in every scenario (maximum absolute residual $0.000B at three decimals). Where the paper compares against the Budget Lab (Section 6), it restates results on their narrower basis: federal individual income tax net of refundable credits, plus payroll.

3.2 Design I: the mechanism sweep

The sweep isolates composition. In 10-point steps from 0% to 100%, it reduces every person's positive employment and self-employment income by the step share and redistributes the same

weighted total to positive capital income in proportion to existing holdings, holding modelled market income fixed. Capital income here means the six directly held flows (long- and short-term capital gains, taxable interest, qualified and non-qualified dividends, and rental income); the calibrated scenarios of Section 3.3 use a wider nine-variable set that adds pass-through and taxable retirement income, which is why the two baselines' capital aggregates differ. The 2026 sweep baseline contains \$12.4T of positive labor income and \$2.7T of positive capital income on the six-variable definition, with 51% of households receiving any positive capital income — so the sweep concentrates income by construction, and the question is what current law does about it.

The sweep's limitation is deliberate: a pure reallocation holds output fixed, while forecasters expect AI to raise output *and* tilt its composition. The sweep therefore answers "what does composition alone do to current law?", and the calibrated scenarios answer "what happens on the paths forecasters actually expect?". The two connect through a forecast-equivalence mapping: the proportional reduction in labor's income share that each tilt produces, $k = 1-(1+g_L)/(1+g_Y)$, is also the sweep position delivering the same share reduction, placing the scenarios at 0.9% (Slow), 2.7% (Moderate) and 6.3% (Rapid) — the sweep's forecast-relevant range is its first decade, and the rest is out-of-sample stress test.

3.3 Design II: forecaster-calibrated scenarios

The scenarios adopt the Budget Lab's calibration of Karger et al. (2026) exactly, as revised on 9 October 2026. Each scenario is defined by an annual real GDP growth rate under AI (Slow 2.0%, Moderate 2.6%, Rapid 3.3%) and a 2030 labor share of factor income (55.0%, 54.0%, 52.0%), both the medians of the survey's economist respondents. The report as first published took the labor shares from the pooled all-respondent column instead (53.8% for Moderate, 51.3% for Rapid); its revision, and this version of the paper, use the economist column for both inputs. Growth is expressed in excess of the CBO baseline path: the five-year output shock is $g_Y = (1+r_{\text{ai}})^5/G_{\text{CBO}} - 1$, and the factor shocks follow from the target shares,

$$g_K = \frac{\theta_K^1(1+g_Y)-\theta_K^0}{\theta_K^0}, \qquad g_L = \frac{\theta_L^1(1+g_Y)-\theta_L^0}{\theta_L^0}. \tag{1}$$

The two remaining inputs are the cumulative CBO baseline factor, $G_{\text{CBO}} = 1.022 \times 1.018^4 \approx 1.0976$ from the CBO growth path their report states (2.2% in 2026, 1.8% a year after), and the pre-shock labor share $\theta_L^0 = 0.555$, which their model repository commits. With these, unit tests reproduce their committed per-cell output, capital and labor growth rates and λ to machine precision in both versions of their release, which pins both constants, and reproduce the rounded Table 1 of both versions to within 0.005 pp.

The wage-inequality channel is their stylization $\lambda = 1 \pm g_Y$: a spread transform on log labor income, $\ln Y_i^1 = \mu^1 + \lambda\,(\ln Y_i^0 - \mu^0)$, equivalently $Y_i^1 = C\,Y_i^{0\,\lambda}$ with C in closed form, with the aggregate pinned to $(1+g_L)$ times baseline. Compressive ($\lambda < 1$) narrows the wage distribution around the same aggregate; expansive ($\lambda > 1$) widens it; proportional scales every wage equally. The transform is applied at the person level because the payroll result turns on a per-person threshold, the Social Security taxable maximum (\$214,200 in 2030; at baseline 15.8% of wage income and 5.7% of workers sit above it).

The capital shock scales nine capital income variables — long- and short-term capital gains, taxable interest, qualified and non-qualified dividends, rental income, partnership and S-corporation income, taxable pension income and taxable IRA distributions — in proportion

to existing holdings, after routing pass-through income through the Budget Lab's labor/capital split rule. The wage-spread transform is not applied to the labor portion of pass-through income, which follows the pass-through rule instead. Appendix B traces every shocked variable into IRS gross income and household market income, confirming the shock reaches the tax base rather than being recomputed away; the labor aggregate lands on its target to within relative error 2×10^{-16}. Each scenario also has a shares-fixed counterfactual growing both factors at g_Y, the comparison behind the Budget Lab's finding that revenue gains would be "nearly twice as large" with factor shares held fixed ("roughly twice" before the revision).

Two further axes vary definitional assumptions their report flags. A *realization rate* routes a fraction of the incremental capital flow outside the taxable base (their assumption corresponds to 1.0); and a *capital-scope* variant excludes taxable pension income and IRA distributions from the capital set, since those are taxed at ordinary rates and their inclusion works against the "capital is taxed lightly" mechanism.

4 Results

4.1 The mechanism at full range

Figure 1 runs the composition mechanism from 0% to 100%. Net government revenue falls almost monotonically as labor income becomes capital income, bottoming at -880B at a 90% shift. The decomposition is not what a "capital-is-taxed-lightly" intuition alone would predict: the federal income tax *gains* at every step (+44B at the first), because concentrating income into the small capital-holding population pushes it into upper brackets by more than the preferential rates give back. Revenue falls anyway because payroll collections fall roughly linearly with the wage base (-147B at the first step) and benefit outlays and refundable credits rise as wage-dependent households qualify for means-tested support; state codes track the federal income-tax pattern. The distributional counterpart (Figure 2) is severe by construction — only 51% of households hold any positive capital income, so the net-income Gini rises from 0.470 to 0.768 and SPM poverty from 13.2% to 61.2% at the full shift, with the top decile's mean net income rising +58% while decile 5 falls -48.5%. The exception is instructive and is the transfer system's signature: the bottom decile ends essentially where it started (+0.8%), because its income is largely transfers, which do not follow wages out of the labor market. The forecast-equivalent range — the shaded band at 0.9–6.3% — says how much of this axis the 2030 forecasts actually reach: the first decade. The sweep's value is the mechanism it exposes, not the tail. These sweep runs predate the correction noted before the introduction: their benefit outlays and net income still include Head Start and Early Head Start.

4.2 Calibrated scenarios

Table 1 reports all twelve scenario cells at 2030 against the same-year current-law baseline; Figure 3 plots the Rapid pattern. Two published Budget Lab mechanisms reproduce on independent machinery. Within each scenario, moving from compressive to expansive raises income-tax revenue through progressivity and lowers payroll revenue as earnings cross the Social Security taxable maximum. And the tilt is expensive: Table 2 shows each scenario keeping a fraction of its shares-fixed revenue — 47% in Rapid (+272B against +576B) on this all-government household basis, against 58% on the narrower federal basis Section 6 matches to the Budget Lab. The difference between those two keep-rates is mainly the Budget Lab's corporate wedge, which

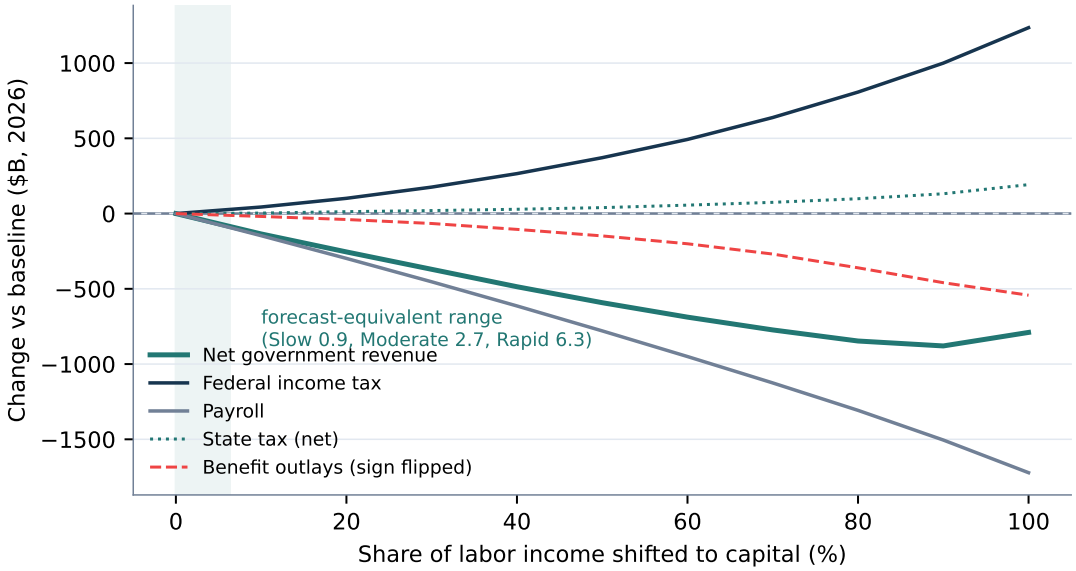


Figure 1: Revenue under a 0–100% labor-to-capital shift, 2026. Lines are changes against baseline; benefit outlays are sign-flipped so a rising line is a fiscal cost. The shaded band marks the forecast-equivalent positions of the 2030 scenarios.

the federal basis includes: it grows with the capital shock, so it cushions the as-forecast cell. The state and transfer channels add to the tilt's dollar cost but leave the keep-rate essentially unchanged.

The payroll-cap mechanism runs in both directions, which is worth stating because it is easy to get backwards. Every Rapid variant grows aggregate labor income by the same 0.41%; only its distribution differs. Under compression, payroll receipts *rise* +58B, against +7B when the same growth is spread proportionally, because compression moves earnings from above the taxable maximum to below it; under expansion the same cap turns that wage growth into a payroll loss of -47B. The wage-spread parameter also has a direct household reading: each variant has a crossover wage below which workers lose (expansive, $139,212) or gain (compressive, $142,922), so under Rapid / expansive the great majority of workers sit on the losing side of a scenario that adds +313B to net revenue.

4.3 The transfer system

The Revenue column of Table 1 nets transfers out; pulling them apart shows the safety net responding exactly as designed (Figure 4). Across the Rapid wage-spread variants — identical aggregate labor income, only its distribution varies — gross tax collections swing $123B, and automatic transfer responses absorb $43B of that swing, 35%. Non-credit transfers alone swing $28B: -15B under compressive, as rising low-end wages float households off means-tested programs, to +13B under expansive as falling ones push them on. SNAP carries $19B of the swing (-10B to +9B) and SSI $3B; Moderate shows the same shape (-8B to +6B). Refundable credits contribute the remaining $15B — a response the Budget Lab's accounting does capture through its tax-credit outlays line. The $28B of program spending is the piece outside their model's coverage, and it is the outlay side of the 3.23pp poverty span: the same variant uncertainty



Figure 2: Inequality and poverty under the same sweep.

their report scores as a revenue question arrives at households as a benefits question.

4.4 States

State codes are the other channel a federal model does not carry. Under Rapid / proportional they collect +42B, and the state take nearly doubles across the wage-spread variants (+30B to +54B) as progressive state schedules concentrate collections at the top the way federal brackets do. On the Rapid / expansive cell used for the reconciliation below, state tax net of refundable state credits adds +54B — roughly a sixth on top of the bridged federal take.

The geography is concentrated (Figure 5): the top five states collect 55% of the national state take (CA +11.6, NY +6.4, NJ +2.0, MA +1.5, MD +1.4, in $B). 7 states collect exactly nothing from the capital shock — Florida, New Hampshire, Nevada, South Dakota, Tennessee, Texas and Wyoming — and Washington is the instructive exception: it has no income tax either, but its capital-gains excise tax collects +1.1B while the seven zero-collecting states collect nothing. A state that taxes capital gains participates in an AI capital boom; a state that taxes only wages does not participate in an economy where wages are the shrinking share. State-funded benefit responses are negligible in these runs (the model's state-benefit registry at version 1.764.6 covers state SSI supplements and child-care subsidies), so the automatic-stabilizer response above is almost entirely federal. Sub-national cells this small carry material data-calibration sensitivity — Section 5.3 shows a single data-build revision moving New York's take from $3.8B to $5.8B — so the state figures locate the pattern rather than measure any one state precisely.

5 The two assumptions that dominate

5.1 Realization

The Budget Lab's capital shock assumes AI leaves the realized share of capital income unchanged, and they flag the assumption as a limitation. It carries real weight: unrealized gains, retained earnings and accruals inside retirement accounts never reach the individual tax base. Holding Rapid / proportional fixed and varying only how much of the incremental capital flow

Scenario	Revenue	Income tax	Payroll	State	Poverty	Δpp	Net Gini
Slow / shares fixed	+47	+29	+10	+6	11.99%	-0.14	0.4680
Slow / compressive	+3	+3	-1	+1	12.01%	-0.12	0.4675
Slow / proportional	+7	+11	-5	+2	12.17%	+0.04	0.4685
Slow / expansive	+10	+19	-10	+3	12.33%	+0.20	0.4695
Moderate / shares fixed	+287	+178	+61	+35	11.46%	-0.67	0.4690
Moderate / compressive	+141	+72	+39	+17	11.11%	-1.02	0.4644
Moderate / proportional	+161	+121	+13	+22	11.89%	-0.24	0.4706
Moderate / expansive	+182	+172	-13	+29	12.77%	+0.64	0.4769
Rapid / shares fixed	+576	+358	+121	+71	11.04%	-1.09	0.4703
Rapid / compressive	+233	+124	+58	+30	10.49%	-1.64	0.4620
Rapid / proportional	+272	+220	+7	+42	11.96%	-0.17	0.4742
Rapid / expansive	+313	+322	-47	+54	13.72%	+1.59	0.4867

Table 1: Scenario grid, 2030, $B change against the current-law baseline (SPM poverty 12.13%, net Gini 0.4678). "Revenue" is net government revenue (federal and state taxes minus refundable credits and benefits); "Income tax" is federal income tax gross of refundable credits; "State" is state tax before refundable state credits.

Scenario	Shares fixed	As forecast	Share kept
Slow	+47	+7	14%
Moderate	+287	+161	56%
Rapid	+576	+272	47%

Table 2: What the tilt toward capital costs: net revenue with factor shares fixed versus as forecast (proportional wage spread), $B, 2030.

lands on a tax return (Figure 6, Table 5 in the appendix), revenue runs from +26B at a 0% realized share to +272B at their 100% assumption — linear to within a percent across a grid in 25-point steps (no point lies more than 0.6% of the range off the straight line through the ends). Revenue stays positive across the whole grid: Rapid labor income grows 0.41% on the revised inputs, so the household side gains even when none of the incremental capital income reaches a return. On the Budget Lab's July inputs, under which Rapid labor income fell, the same sweep turned negative below a 22.5% realized share.

One counterweight sits outside the household sector: corporate tax is levied upstream of household realization, so the Budget Lab's corporate wedge would still collect about +74B on the same shock regardless of the realized share, lifting the all-in total (household, state and corporate) to about +101B at the 0% floor. Low realization shifts the fiscal gain from an AI boom toward the corporate side; it does not eliminate it. One scope note: the realization haircut applies uniformly to all nine capital variables, including pass-through income and retirement distributions that are realized by construction, so the low end of the axis is a stress bound rather than a describable world.

Poverty is flat across the entire sweep, between 11.96% and 12.03%. Whether AI's capital gains are taxable moves net revenue by roughly a quarter-trillion dollars ($245B) and moves poor households essentially not at all: the transfer system responds to what happens to wages, not to how much of the capital gain the tax system reaches.

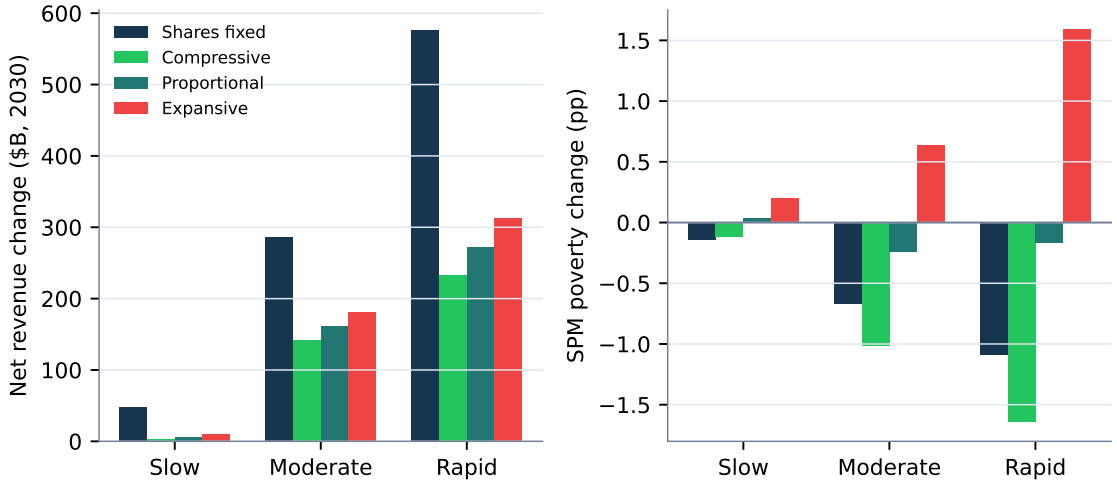


Figure 3: Net revenue (left) and SPM poverty change (right) by scenario and wage-inequality variant, 2030.

5.2 What counts as capital

Taxable pension and IRA distributions are taxed at ordinary rates. Counting them in the capital set — which the Budget Lab does, and the headline results follow — raises the revenue estimate and softens the "capital is taxed lightly" mechanism. Excluding them (Table 6 in the appendix) moves Rapid from +272B to +177B and flips Slow's sign (+7B to -4B). Realization alone spans $245B of the Rapid estimate, nearly the $265B that separates the Slow and Rapid headlines; the retirement-scope choice, worth $95B, falls inside that range.

5.3 Stability across data builds

The results were first computed on the prior `populace-us` build (build o, 22 July 2026) and rerun on build p (28 July 2026, with revised capital gains calibration) under the identical pinned model version — a data-only change. Both runs predate the Head Start correction and the October revision noted before the introduction, so this comparison uses the net-income accounting and the July scenario inputs they shared. No scenario revenue figure moved by more than $5.9B in absolute terms; in relative terms Moderate and Rapid cells moved by at most 2.8%, while the smallest Slow cells — single-digit-billion bases — moved by up to 12%. Every poverty change moved by less than 0.11 pp, and no conclusion changed. The recalibration's visible effect is distributional: the baseline top-1% net income share rises from 8.77% to 9.00% (9.06% after the correction), and New York's state take under Rapid / proportional rises from $3.8B to $5.8B as more of the gains land where they are taxed. This build-to-build movement bounds sensitivity to data calibration, not sampling error, which is unquantified here (Section 7). An average-tax-rate cross-check points the same way as the reconciliation. The Budget Lab's Rapid band, 21% to 32% across the wage-spread variants, is federal revenue including their corporate wedge over the gross factor income their shock adds ($767B); their cells run 21/26/32% for compressive / proportional / expansive. On the same numerator — federal income tax net of credits, payroll and their wedge — over the market income the shock adds here (+887B under proportional), this paper's cells run 30/34/39%, higher in every cell, as the bridged headline is.

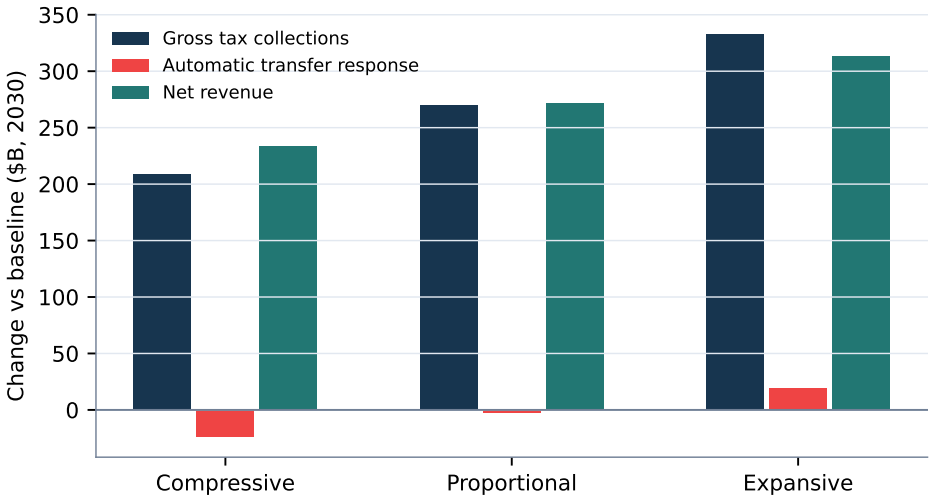


Figure 4: Gross tax collections, automatic transfer response, and net revenue across the Rapid wage-inequality variants, 2030. Transfers (refundable credits plus benefits) rise exactly when workers lose, absorbing 35% of the gross-tax swing across the variants.

The denominators differ, so the comparison is indicative.

6 Reconciliation with the Budget Lab

The two models measure different totals: theirs is federal individual income tax plus payroll plus a corporate wedge, with no states and no non-credit transfers; this paper's household total spans federal and state and nets out refundable credits and the benefit programs counted in household net income, a set that excludes health programs, Head Start and Early Head Start (Section 3). Setting the headline numbers side by side without a bridge would be wrong, so this section builds one, using only this paper's run output and objects they publish. Their repository commits the complete published result grid — revenue by instrument for all eighteen cells — which makes the comparison possible at the cell level rather than against two text-stated totals.

Basis. Their microsimulation total is individual income tax plus payroll minus tax-credit outlays; the matching quantity here is federal income tax net of refundable credits, plus payroll (both halves and self-employment). Their corporate wedge reduces to the capital growth rate times CBO's 2030 corporate anchor of $477.3B — in their formula $\Delta R_{\text{CIT}} = X \cdot (R_{\text{CIT}}/Y_K^0)$ with $X = Y_K^0\, g$, the capital base cancels — and all 18 committed values equal $g \times \$477.3\text{B}$ to within $0.000B, checked against their own per-cell capital growth rates. Adding their wedge to this paper's federal microsimulation total yields a like-for-like "bridged" headline. The wedge is not PolicyEngine modelling corporate tax, and it inherits every constant-rate assumption they list.

Where the models agree. Payroll responses agree closely in all nine as-forecast cells — the largest gap anywhere in the grid is $5.1B (Table 4). Across the Rapid variants ours run +58/+7/-47 against theirs +54/+5/-48 for compressive / proportional / expansive — two independent models putting similar wage distributions against the same taxable maximum, including the sign reversal under expansion, where aggregate labor income grows but payroll

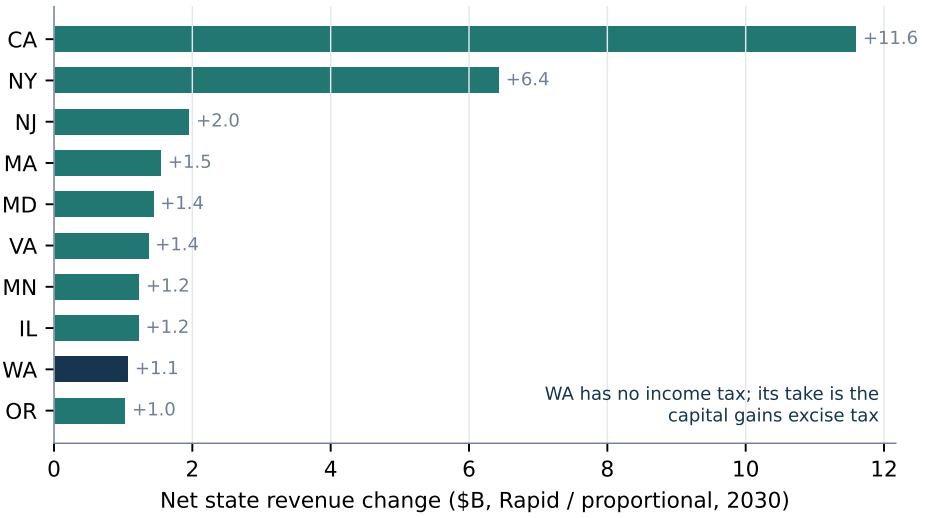


Figure 5: Top ten states by net state revenue change, Rapid / proportional, 2030.

Rapid / expansive	IIT (net)	Payroll	Corporate wedge	Total
This paper	+316	-47	+74	+344
Budget Lab	+220	-48	+74	+246

Table 3: The bridged headline cell (Rapid / expansive, their headline scenario and the largest gain they report), $B, 2030. "IIT (net)" is individual income tax net of refundable credits; the corporate wedge is theirs in both rows.

receipts fall. The tilt ratios also line up once stated on the same basis: on their federal-plus-corporate accounting, shares-fixed revenue exceeds as-forecast revenue by 3.16× / 1.53× / 1.72× here against 3.68× / 1.58× / 1.79× in their grid for Slow / Moderate / Rapid — agreement on the paper's central mechanism, with the same ordering. (The household-total ratio for Rapid is larger, 2.12×, mainly because the household basis omits the corporate wedge: the wedge grows with the capital shock, so it cushions the as-forecast denominator on the matched basis. States and transfers raise the tilt's dollar cost but leave the ratio essentially unchanged.) Their most extreme published comparison reproduces in direction but not in magnitude: for Slow with compressive inequality, their grid puts revenue 84% below the shares-and-distribution-unchanged counterfactual (their text states 82%), while this paper gets 93%. Slow cells are also the least stable in this model: Section 5.3 shows them moving up to 12% on a data-build refresh alone, because their bases are single-digit billions. The agreement to take seriously in this scenario is the sign and the ordering, not the second digit.

Where they diverge, and why. The bridged headline is +344B against their +246B — 1.39×, or 5.2% against 3.7% of the CBO 2030 revenue baseline ($6,595B, from their committed grid). The gap concentrates in the income-tax response to the capital shock, and the absolute gap says so more clearly than any ratio: it is nearly constant at +98/+100/+96 $B across the three Rapid wage-spread cells, exactly what a divergence in the shared capital shock — rather than in the varying wage channel — would produce. (In ratio terms: 1.82× theirs on

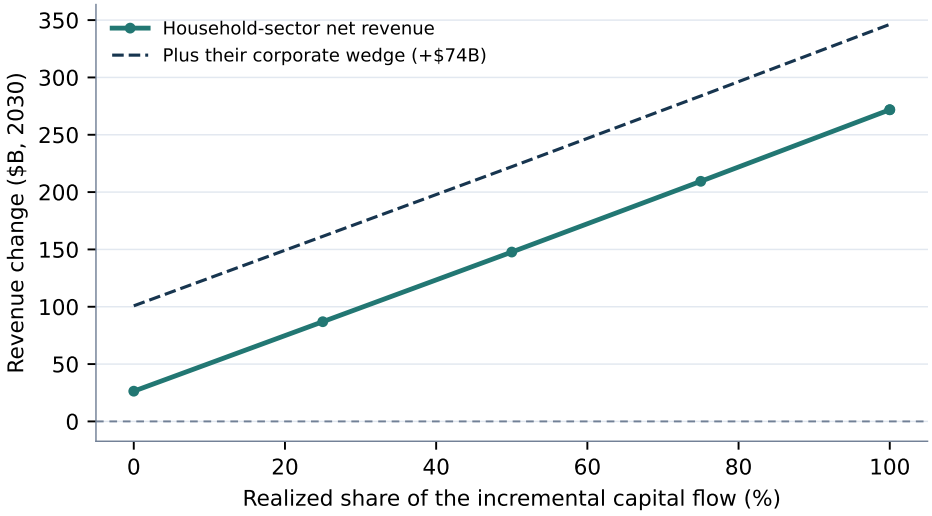


Figure 6: Net revenue under Rapid / proportional as the realized share of the incremental capital flow varies. The dashed line adds the Budget Lab's realization-independent corporate wedge.

Cell	IIT net of credits		Payroll	
	Ours	Theirs	Ours	Theirs
Slow / compressive	+3	-4	-1	+0
Slow / proportional	+11	+4	-5	-4
Slow / expansive	+18	+11	-10	-8
Moderate / compressive	+77	+25	+39	+34
Moderate / proportional	+122	+70	+13	+9
Moderate / expansive	+169	+118	-13	-17
Rapid / compressive	+132	+33	+58	+54
Rapid / proportional	+221	+121	+7	+5
Rapid / expansive	+316	+220	-47	-48

Table 4: Cell-level comparison on the matched federal basis, all nine as-forecast cells, $B, 2030.

the proportional cell, where the wage-spread channel is neutral; 1.44× on the expansive bridge cell.) Three model differences point the same direction. First, the realized-capital base: their committed cell parameters put their baseline realized taxable capital base at $4.51T, against $5.3T here — 17% larger. Second, allocation: they distribute the incremental capital flow by SCF-imputed total wealth; this paper distributes by existing realized capital income, which they note is the more top-concentrated choice — and top-concentrated dollars face higher marginal rates. Third, retirement routing: taxable pension and IRA distributions here scale with the capital shock and are taxed at ordinary rates. The capital-scope sensitivity of Section 5.2 shows that last channel alone is worth about $95B in Rapid.

7 Discussion

7.1 What the results say

Three readings of the results bear directly on policy questions the AI-fiscal conversation is starting to ask. First, the fiscal and household questions decouple. Capital-side tax design (the realization margin) moves the budget by roughly a quarter-trillion dollars while leaving poverty untouched; the wage-side distribution moves poverty by 3.23pp while net revenue stays positive throughout. A revenue estimate, however good, is not a household forecast. Second, the automatic stabilizers are already in the scenario arithmetic: transfers absorb 35% of the gross-tax swing across the wage-spread variants, which means deficit projections that stop at the tax code overstate how much the wage-distribution assumption matters fiscally — and understate the program-side costs when workers lose. Third, the state layer is not a rounding error: +42B under proportional Rapid growth, nearly doubling across wage-spread variants, concentrated in a handful of states, and available only to states whose codes reach capital income at all.

7.2 Limitations

No corporate sector. PolicyEngine-US models households; the corporate wedge in Section 6 is borrowed from the Budget Lab, not modelled.

Static. No labor supply, saving, avoidance or enforcement responses on either side of the comparison.

Apportionment is narrower than theirs. The incremental capital flow follows existing realized capital income; their SCF-wealth apportionment reaches households that hold wealth but realize nothing. By their own account the narrower base is more top-concentrated, so this choice concentrates the shock more than theirs. The model carries household balance-sheet variables (`net_worth`, `stock_assets`), so an asset-based variant is feasible and is identified future work; the open modelling question is which person and which income type receives the new flow in a household that currently realizes none.

Pass-through split is coarser than theirs. Their active/passive-aware rule is approximated by a single rule because the microdata carries no active/passive flag, under-assigning the passive portion to capital.

National shares versus modelled shares. Scenario calibration uses national factor shares ($\theta_L^0 = 0.555$); observed taxable microdata flows are more labor-weighted ($15.9T labor against $5.3T capital), because survey and tax data miss unrealized gains, retained earnings and imputed returns. Growth rates from national accounts are applied to observed taxable flows — the same choice the Budget Lab makes with the PUF — and both baseline shares are reported so the gap is visible rather than assumed away. One implication is worth stating for the headline tilt result: under Rapid, the as-forecast scenario raises modelled market income by +887B while the shares-fixed counterfactual raises it by +1519B, because much of the forecast capital growth accrues to income the taxable microdata does not observe. The tilt's revenue cost therefore combines a composition effect with growth landing outside the modelled base; the realization sweep of Section 5.1 is the sensitivity that spans this margin.

No sampling uncertainty. Estimates carry survey-sampling and imputation error that is not quantified here. The build-to-build comparison of Section 5.3 bounds sensitivity to data calibration — material for sub-national and program-level cells — but not sampling error; small cells such as individual state takes and single-program responses should be read to the nearest billion, not the decimal shown.

Anchored poverty, unequivalized Gini. SPM thresholds do not respond to the simu-

lated distribution, so a fully relative poverty line would rise with five years of above-baseline growth and show larger increases. The Gini is household-level and unequivalized: changes are meaningful, levels are not comparable to published equivalized series. Poverty levels sit above the published Census SPM rate (Shrider and Bijou, 2025) — a known calibration gap — so every poverty figure is a change against a same-year baseline on identical thresholds.

No health-program response. Medicaid, CHIP and ACA subsidies are excluded from net-income accounting in these runs by the model's default switch; the transfer responses reported here would not shrink, and could grow, with that channel on — but this paper does not measure it.

7.3 Next steps

The wage-inequality parameter λ is a stylization with, as the Budget Lab says plainly, little evidence behind its magnitude. The natural replacement is measured occupation-level exposure (Felten et al., 2021; Eloundou et al., 2024): distributing the labor shock by exposure scores mapped to CPS occupations would make the implied λ an output rather than an input — the design of the UK companion paper (Ahmadi, 2026), and the "S1" expansion the Budget Lab's methodology names as planned future work. Asset-based capital apportionment and a policy-response layer (recycling the net revenue gain into transfers to test whether the receipts can hold the bottom of the distribution harmless) are the other two identified extensions.

7.4 Conclusion

Whether AI pays for itself fiscally depends on where its income lands — between factors, across the wage distribution, and on or off the individual tax base. Running forecaster-calibrated scenarios through the full United States tax and transfer system, the tilt toward capital keeps 58% of the revenue that balanced growth would deliver on the federal basis and 47% on the household basis (states and transfers, no corporate wedge), and it erodes most of growth's poverty gain; the wage-distribution assumption nobody can yet pin down flips the household verdict while the revenue verdict stands; and the channels a tax-only model omits — automatic transfers and state codes — carry a material share of the response. Where the two models share machinery — payroll and the tilt ratio — they agree closely; they diverge on the income-tax response to the capital shock, a gap that traces to a larger capital base, a more top-concentrated allocation rule and retirement routing. All of it is open: anyone can rerun these scenarios, or score a different future against them.

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A Additional tables

Realized share	Revenue
0%	+26
25%	+87
50%	+148
75%	+209
100% (their assumption)	+272

Table 5: Net revenue under Rapid / proportional by realized share of the incremental capital flow, $B, 2030.

Scenario	Full capital set	Excl. retirement	Difference
Slow	+7	-4	-10
Moderate	+161	+118	-43
Rapid	+272	+177	-95

Table 6: Capital-scope sensitivity: net revenue with and without taxable pension and IRA distributions in the capital set (proportional variants), $B, 2030.

State	Net revenue change
CA	+11.59
NY	+6.43
NJ	+1.95
MA	+1.54
MD	+1.43
VA	+1.36
MN	+1.23
IL	+1.22
WA	+1.07
OR	+1.02
7 states collect exactly $0: FL, NH, NV, SD, TN, TX, WY.	

Table 7: Top ten states by net state revenue change (state tax before refundable credits minus refundable state credits), Rapid / proportional, $B, 2030.

Shift	Revenue	Income tax	Poverty	Net Gini
0%	+0	+0	13.2%	0.470
10%	-135	+44	15.0%	0.490
20%	-255	+101	16.9%	0.513
30%	-370	+176	20.1%	0.536
40%	-486	+265	23.9%	0.561
50%	-592	+371	29.3%	0.588
60%	-688	+493	35.4%	0.617
70%	-773	+638	43.1%	0.647
80%	-847	+807	50.7%	0.681
90%	-880	+1000	57.6%	0.719
100%	-790	+1235	61.2%	0.768

Table 8: The mechanism sweep, 2026: net revenue, federal income tax, SPM poverty and net Gini by share of labor income shifted to capital.

B Shock routing into the tax base

Several shocked variables are aggregates rather than raw inputs, so the implementation verifies that overriding them moves taxable income rather than being recomputed away. Traced against the model's gross-income and market-income source registries (`gov.irs.gross_income.sources` and `gov.household.market_income_sources` in policyengine-us 1.764.6):

Shocked variable	Route into IRS gross income	Route into market income
employment income	via employment income, net of pre-tax contributions	listed source
self-employment income	listed source	listed source
long/short-term capital gains	capital gains	capital gains
qualified/non-qualified dividends	ordinary dividend income	dividend income
taxable interest	listed source	interest income
rental income	listed source	listed source
partnership and S-corp income	listed source	listed source
taxable pension income	listed source	pension income
taxable IRA distributions	taxable retirement distributions	retirement distributions

Table 9: Every shocked variable reaches both the federal tax base and household market income. Qualified and non-qualified dividends are the two components of ordinary dividends; short- and long-term gains the two components of capital gains; tax-exempt interest is deliberately outside the shocked set.

Because every shocked variable reaches household market income, the net-income identity closes; the maximum residual across scenarios is \$0.000B, and the labor aggregate lands on its target to within relative error 2×10^{-16}.

C Reproducibility

Everything in this paper runs from open source against openly published microdata:

- **Model:** policyengine-us 1.764.6 via policyengine.py 5.0.1 (the runtime certifies the model–data pairing).
- **Data:** `populace-us` certified build `populace-us-2024-buildp-sparse-rmloss100-cae8640-20260728T0`
- **Code:** `github.com/PolicyEngine/ai-inequality` — scenario engine in `analysis/ai_scenarios.py`, runner in `analysis/compute_ai_scenarios.py`, sweep in `analysis/compute_shift_sweep.py`, reconciliation in `analysis/reconcile_budget_lab.py`. Every number and table in this paper is emitted from the run outputs by `analysis/emit_paper_values.py`; figures by `analysis/paper_figures.py`.
- **Their side:** the Budget Lab's `AI-Fiscal` repository (MIT license) commits the complete published result grid (`results/aggregates/ai_fiscal_publishable_2030_latest.xlsx`, here at their revised release v1.0.1); the reconciliation reads it directly, and a copy is committed alongside this paper's own run outputs so the bridge is reproducible without a network fetch. Their PUF-based input microdata cannot be redistributed, so their pipeline is readable but not rerunnable by third parties; this paper's is both.
- **Canonical run outputs:** `analysis/outputs/ai_scenarios.json` (build p, the Budget Lab's revised inputs) is committed with five comparison files: `ai_scenarios_v1_0_0_inputs.json` (the same run on their July inputs, the basis of the revision and correction notes) and `yale_publishable_2030_v1_0_0.xlsx` (their July result grid); `ai_scenarios_buildp_published.json` (build p as first published, with Head Start and Early Head Start in net income) and `ai_scenarios_buildo.json` (build o), the stability comparison of Section 5.3; and `transfer_detail_buildp` program-by-program benefit totals on the published runtime, the source of the Head Start amounts in the correction note. Every value and table in this paper regenerates from the repository alone — the full simulation need only be rerun to change a scenario, not to check a number.
- **Interactive companion:** https://policyengine.org/us/ai-inequality/income-shift (scenario explorer) and https://policyengine.org/us/ai-inequality/budget-lab (memo-length summary).

Calibration unit tests reproduce the Budget Lab's committed per-cell growth rates and λ to machine precision, and all nine published derived values of their Table 1 to within 0.005 pp, in both the July and the revised October version of their release, from the CBO path their report states ($G_{\text{CBO}} = 1.022 \times 1.018^4$) and the pre-shock labor share their repository commits ($\theta_L^0 = 0.555$). The scenario runner checkpoints per-scenario results and is resumable; run logs and build-o archives are kept under `analysis/outputs/`.